Algebraic Topology

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1 Homology

If we want an invariant that detects higher dimensional holes, the natural choice would be to consider $\pi_n(X)$ the homotopy classes of maps $S^n \to X$; unforutnately, these are very difficult to compute. By adding some combinatorial structure to our shapes, and keeping our groups abelian, we can define a functor that's much easier to compute on many of the nice spaces we care about.

1.1 Simplicial homology

Definition 1.1.1. A **Δ-complex** structure on a space X is a collection of maps:

$$\sigma_\alpha : \Delta^n \to X$$

where n depends on α, with all of the σ_α satisfying the following:

- The restriction of each σ_α to $\mathring{\Delta}^n$ is injective;
- Each point $x \in X$ lies in exactly one of the $\sigma_\alpha(\mathring{\Delta}^n)$;
- For each face map δ_i, there exists some σ_β in the Δ-complex with $\sigma_\alpha \circ \delta_i = \sigma_\beta$;
- A subset $U \subset X$ is open iff all of the $\sigma_\alpha^{-1}(U)$ is open in each Δ^n.

Δ-complexes are a special example of CW-complexes, thus are Hausdorff. When considering these examples, it is paramount to conflate the most injective maps σ_α with their images in X, which will be simplices with some collapsing of their boundaries.

TODO:

Definition 1.1.2. Given a Δ-complex structure on X, we form the **simplicial complex**:

$$\cdots \xrightarrow{d_{n+1}^\Delta} C_n^\Delta(X) \xrightarrow{d_n^\Delta} C_{n-1}^\Delta(X) \xrightarrow{d_{n-1}^\Delta} \cdots \xrightarrow{d_3^\Delta} C_2^\Delta(X) \xrightarrow{d_2^\Delta} C_1^\Delta(X) \xrightarrow{d_1^\Delta} C_0^\Delta(X) \xrightarrow{d_0^\Delta} 0$$

where $C_n^\Delta(X)$ is the free abelian groups on the n-simplices in X, and the differential is given by

$$d_n^\Delta(\sigma_\alpha) = \sum_{i=0}^{n} (-1)^i \sigma_\alpha \circ \delta_i$$

Proposition 1.1.3. $d_\bullet^\Delta$ does in fact make $C_\bullet^\Delta$ a chain complex, i.e. $d_n^\Delta \circ d_{n+1}^\Delta = 0$ for all n.

Proof. By linearity, it suffices to consider an $(n+1)$-simplex $\sigma_\alpha : \Delta^{n+1} \to X$, on which we have:

$$d_n^\Delta d_{n+1}^\Delta(\sigma) = d_n^\Delta\left(\sum_{i=0}^{n+1}(-1)^i\sigma\delta_i\right) = \sum_{i=0}^{n+1}(-1)^i d_n^\Delta(\sigma\delta_i) = \sum_{i=0}^{n+1}(-1)^i\left(\sum_{j=0}^{i-1}(-1)^j\sigma\delta_i\delta_j - \sum_{j=i+1}^{n+1}(-1)^j\sigma\delta_i\delta_j\right)$$

in this sum, for each $0 \leq a < b \leq n+1$, there will be two terms of $\sigma\delta_a\delta_b$, one with coefficient $(-1)^{a+b}$, and the other with $-(-1)^{a+b}$, thus the sum is zero on all $(n-1)$-simplices. $\square$

Definition 1.1.4. If X has a Δ-complex structure, then we define the **simplicial homology** of X, as the homology of the chain complex $(C_\bullet^\Delta, d_\bullet^\Delta)$, we write this as $H_*^\Delta(X) := H_*(C_\bullet^\Delta(X), d)$.

TODO: computations and pictures

Maps of spaces don't necessarily induce maps on chain complexes, there is no guarantee that the image of the Δ-complex is a Δ-complex.

TODO: give an example

1.2 Singular homology

We want a formulation of homology that is immediately functorial.

Definition 1.2.1. To a space X we can associate the **singular complex** $C_\bullet(X)$, when $C_n(X)$ is the free abelian group on maps $\sigma : \Delta^n \to X$, and the differential is as with the simplicial complex.

From this we also define the **singular homology** of X, $H_*(X) := H_*(C_\bullet(X), d)$

Proposition 1.2.2. A continuous map $f : X \to Y$ induces homomorphisms $f_* : H_n(X) \to H_n(Y)$ for all n, this makes each homology group a functor $H_n(-) : \mathbf{Top} \to \mathbf{Ab}$.

Proof. We will show f induces a chain map $f_\sharp : C_\bullet(X) \to C_\bullet(Y)$. We define $f_\sharp$ on generators by $f_\sharp(\sigma) = f \circ \sigma$, to see this is a chain map observe:

$$f_\sharp(d\sigma) = f_\sharp\left(\sum_{i=0}^{n}(-1)^i\sigma \circ d_i\right) = \sum_{i=1}^{n}(-1)^i f_\sharp(\sigma \circ d_i) = \sum_{i=1}^{n}(-1)^i(f \circ \sigma) \circ d_i = d(f \circ \sigma) = d(f_\sharp)$$

All the other properties are obvious. $\square$

Corollary 1.2.3. If $f : X \to Y$ is a homeomorphism, then f_* is an isomorphism.

We can go further, showing homotopy equivalences induce isomorphisms on homology.

Proposition 1.2.4. If $f, g : X \to Y$ are homotopic maps, then the chain maps $f_\sharp, g_\sharp$ are chain homotopic,[1] thus $f_* = g_*$.

Proof. Let $H : X \times I \to Y$ be the homotopy from f to g, then $H \circ (\sigma \times \mathrm{id})$ is a homotopy from $f \circ \sigma$ to $g \circ \sigma$. But the domain of this map, $\Delta^n \times I$, isn't a simplex, we can decompose it as:

$$\Delta^n \times I = [v_0, \ldots, v_n] \times I = \bigcup_{i=0}^{n}[(v_0, 0), \ldots, (v_i, 1), (v_i, 1), \ldots, (v_n, 1)] =: \bigcup_{i=0}^{n} G_\sigma^i$$

TODO: low dimensional picture

Giving us $n+1$ n-simplices in Y associated to σ, let's defien the obvious maps $K_n : C_n(X) \to C_{n+1}(Y)$:

$$K_n(\sigma) = \sum_{i=0}^{n}(-1)^i G_\sigma^i$$

which I claim is a chain homotopy. Observe:

$$dK_n(\sigma) = \sum_{i=0}^{n}(-1)^i d(G_\sigma^i) = \sum_{i=0}^{n}\left(\sum_{j=0}^{i}(-1)^{i+j}G_\sigma^i \circ \delta_j - \sum_{j=i}^{n}(-1)^{i+j}G_\sigma^i \circ \delta_{j+1}\right)$$

$$K_{n-1}(d\sigma) = \sum_{i=0}^{n}(-1)^i K_{n-1}(\sigma \circ \delta_i) = \sum_{i=0}^{n}\left(\sum_{j=0}^{i-1}(-1)^{i+j}G_\sigma^j \circ \delta_{i+1} - \sum_{j=i+1}^{n}(-1)^{i+j}G_\sigma^j \circ \delta_i\right)$$

[1] Two chain maps $f, g : A_\bullet \to B_\bullet$ are called **chain homotopic** if there exists some $K : A \to B[1]$ such that $f - g = dK + Kd$.

with some careful inspection, and some picturing the $n = 2$ case, we can see the only terms remaining in the sum are those where we miss a point with the index that G double counts:

$$dK_n(\sigma) + K_{n-1}d(\sigma) = \sum_{i=0}^{n}(G^i_\sigma \circ \delta_i - G^i_\sigma \circ \delta_{i+1}) = G^0_\sigma \circ \delta_i - G^n_\sigma \circ \delta_{n+1} = g \circ \sigma - f \circ \sigma$$

Thus, by linearity, $K_\bullet$ is a chain homotopy. □

So we can view each singular homology groups as a functor

$$H_n : \mathbf{hTop} \to \mathbf{Ab}$$

from the category of topological spaces and *homotopy equivalence classes* of continuous maps.

Corollary 1.2.5. If $f : X \to Y$ is a homotopy equivalence, then $f_* : H_n(X) \to H_n(Y)$ is an isomorphism.

1.3 Relative homology

Lemma 1.3.1. Given a short exact sequence of chain complexes:

$$0 \to A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \to 0$$

there is a long exact sequence of homology groups:

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Proof. We first define ∂:

$$\begin{tikzcd}
0 \arrow[r] & A_n \arrow[r, "i"] \arrow[d, "d"] & B_n \arrow[r, "j"] \arrow[d, "d"] & C_n \arrow[r] \arrow[d, "d"] & 0 \\
0 \arrow[r] & A_{n-1} \arrow[r, "i"] \arrow[d, "d"] & B_{n-1} \arrow[r, "j"] \arrow[d, "d"] & C_{n-1} \arrow[r] \arrow[d, "d"] & 0 \\
0 \arrow[r] & A_{n-2} \arrow[r, "i"] & B_{n-2} \arrow[r, "j"] & C_{n-2} \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
 & b \arrow[r, mapsto] \arrow[d, mapsto] & c \arrow[d, mapsto] \\
a \arrow[r, mapsto] \arrow[d, mapsto] & d(b) \arrow[r, mapsto] \arrow[d, mapsto] & 0 \\
0 \arrow[r, mapsto] & 0 &
\end{tikzcd}$$

Following the above diagram: given some $[c] \in H_n(C)$, as j is surjective there is some $b \in B_n$ with $j(b) = c$, by commutativity we have $j(d(b)) = d(c) = 0$, so $d(b) \in \ker j = \operatorname{im} i$, as i is injective we can find a unique $a \in A_{n-1}$ with $i(a) = d(b)$, once again by commutativity $i(d(a)) = d(d(b)) = 0$, as i is injective we must have $d(a) = 0$ giving a homology class $[a] \in H_n(A)$ which take as $\partial([c])$.

We have to check our choice is well defined, as we have made a choice for b; if we chose a different $b' \in B_n$ with $j(b') = c$, then $j(b - b') = 0$, thus $b - b' \in \operatorname{im} i$ with preimage $a' \in A_n$ so $b' = b + i(a')$, now $d(b') = d(b) + di(a') = d(b) + id(a') = i(a + d(a'))$, which will give the same homology class.

I claim ∂ is now obviously a homomorphism $\ker d_n \to H_{n-1}(A)$.

To show this descends to a map on homology we have to show the image of $\operatorname{im} d \subseteq C_n$ is 0. If $c = d(c')$ for some $c' \in C_{n+1}$, then by surjectivity there exists some $b' \in B_{n+1}$ with $j(b') = c'$, by commutativity our $b = d(b')$ and so $i(a) = d(b) = 0$, thus $a = 0$.

All of the exactness are very direct diagram chases, everything will work if you follow your nose. □

Corollary 1.3.2. For $A \subset X$ a subspace, write $C_\bullet(X, A)$ for $C_\bullet(X)/C_\bullet(A)$, and $H_*(X, A)$ for the homology groups of this complex, then there is a long exact sequence in **relative** homology:

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{\pi_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Corollary 1.3.3. Given a triple $A \subset B \subset X$, by the third isomorphism theorem there is a long exact sequence in relative homology:

$$\cdots \to H_n(B, A) \xrightarrow{i_*} H_n(X, A) \xrightarrow{\pi_*} H_n(X, B) \xrightarrow{\partial} H_{n-1}(B, A) \to \cdots$$

Lemma 1.3.4 (Five lemma)**.** Given the following commutative diagram of abeliang groups:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\alpha"] & B \arrow[r, "g"] \arrow[d, "\beta"] & C \arrow[r, "h"] \arrow[d, "\gamma"] & D \arrow[r, "k"] \arrow[d, "\delta"] & E \arrow[d, "\varepsilon"] \\ A' \arrow[r, "f'"'] & B' \arrow[r, "g'"'] & C' \arrow[r, "h'"'] & D' \arrow[r, "k'"'] & E' \end{tikzcd}$$

if both rows are exact and $\alpha, \beta, \delta, \varepsilon$ are all isomorphisms, then γ is also an isomorphism.

Proof. We prove this with two different *four lemmas*:

$$(1) \quad \begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\alpha"'] & B \arrow[r, "g"] \arrow[d, "\beta"'] & C \arrow[r, "h"] \arrow[d, "\gamma"] & D \arrow[d, "\delta"] \\ A' \arrow[r, "f'"'] & B' \arrow[r, "g'"'] & C' \arrow[r, "h'"'] & D' \end{tikzcd} \qquad (2) \quad \begin{tikzcd} B \arrow[r, "g"] \arrow[d, "\beta"'] & C \arrow[r, "h"] \arrow[d, "\gamma"'] & D \arrow[r, "k"] \arrow[d, "\delta"] & E \arrow[d, "\varepsilon"] \\ B' \arrow[r, "g'"'] & C' \arrow[r, "h'"'] & D' \arrow[r, "k'"'] & E' \end{tikzcd}$$

For (1) we assume both rows are exact, β, δ are injective, and α is surjectve; and show γ is injective:
Have $c \in C$ such that $\gamma(c) = 0 \in C'$ so $\delta h(c) = h'\gamma(c) = 0$ and as δ is injective, $h(c) = 0$;
so $c \in \ker h = \operatorname{im} g$, there exists $b \in B$ with $c = g(b)$ and $g'(\beta(b)) = \gamma(g(b)) = \gamma(c) = 0$;
so $\beta(b) \in \ker g' = \operatorname{im} f'$, there exists $a' \in A'$ with $f'(a') = \beta(b)$;
α surjective $\implies$ there exists $a \in A$ with $a' = \alpha(a)$ and $\beta(b) = f'(\alpha(a)) = \beta(f(a))$;
β injective $\implies$ $b = f(a)$ so $c = g(b) = f(g(a)) = 0$.
For (2) we assume both rows are exact, β, δ are surjective, and ε is injective; and show γ is surjective:
Have $c' \in C'$; as δ is surjective there is a $d \in D$ with $\delta(d) = h'(c')$;
$\varepsilon(k(d)) = k'\delta(d) = k'h'(c) = 0$; as ε is injective, $k(d) = 0$;
so $d \in \ker k = \operatorname{im} h$, there exists $c \in C$ with $h(c) = d$; so $h'(\gamma(c))\delta(h(c)) = \delta(b) = c'$;
so $c' - \gamma(c) \in \ker h' = \operatorname{im} g'$, there is $b' \in B'$ with $g'(b') = c' - \gamma(c)$;
as β is surjective, there is $b \in B$ with $\beta(b) = b'$; so $\gamma(g(b)) = g'(\beta(b)) = c' - \gamma(c)$;
so $c' = \gamma(g(b)) + \gamma(c) = \gamma(g(b) + c)$.
Therefore γ is both injective and surjective, thus an isomorphism. □

All of the previously established properties of homology also hold for relative homology as a functor from the category of pairs of spaces and continuous functions preservings subspaces to **Ab**.

Corollary 1.3.5. Let $f : (X, A) \to (Y, B)$ be a map of pairs. If any tow of the follows maps are isomorphisms, the third is as well:

$$(f|_A)_* : H_*(A) \to H_*(B) \qquad f_* : H_*(X) \to H_*(Y) \qquad f_* : H_*(X, A) \to H_*(X, B)$$

Proof. Apply the five lemma to the induced maps between the long exact sequences of pairs. □

1.4 Barycentric subdivision

Definition 1.4.1. Let $\mathcal{U} = \{U_i\}$ be a collection of subsets of X whose interiors form an open cover of X. For each n consider

$$C_n^{\mathcal{U}}(X) = \sum_i C_n(U_i) \subseteq C_n(X)$$

equipping this with the usual differential gives us the chain complex $C_\bullet^{\mathcal{U}}(X)$ with the canonical natural map $i : C_\bullet^{\mathcal{U}}(X) \to C_\bullet(X)$ which induces $i_* : H_n^{\mathcal{U}}(X) \to H_n(X)$.

With this construction, we want to approximate a general homology class as a sum of small simplices, each of which lie entirely within one component of an open cover. This will be reified with an isomorphism between the $H_*(X)$ and $H_*^{\mathcal{U}}(X)$ above, and our exact method for replacing our simplices with smaller ones will be *barycentric subdivision.*

We define the barycentric subdivision of the n-simplex $\Delta^n = [v_0, \dots, v_n] \subset \mathbb{R}^{n+1}$ inductively. Let $b = \frac{1}{n+1}(v_0 + \cdots + v_n)$, then for each facet $[w_1, \dots, w_n]$ of Δ^n, consider the n-simplex $[b, w_1, \dots, w_n]$. This collection of $(n+1)!$ sipmlices are the barycentric subdivision of Δ^n.

TODO: some pictures in multiple dimensions of multiple iterations of barycentric subdivision.

To show these simplices get appropriately small, recall this lemma from analysis.

Proposition 1.4.2 (Lebesgue covering lemma)**.** Let $\{U_i\}$ be an open cover of a compact metric space X, there exists some $\delta > 0$ such that for all $x \in X$, the ball $B_\delta(x)$ is wholy contained in one of the U_i.

Proof. As X is compact, wlog we may take $\{U_i\}$ to be finite. So we can consider the function $f : X \to \mathbb{R}$ given by

$$f(x) := \frac{1}{n} \sum_{i=0}^{n} d(x, U_i^c)$$

as f is continuous and defined on a compact set, it must achieve a minimum δ. And, as every x is contained in some U_i, f is never 0, thus $\delta > 0$. Now, for all $x \in X$, there exists some U_i such that $d(x, U_i^c) \geq \delta$, thus $B_\delta(x) \subseteq U_i$. □

Lemma 1.4.3. Let $\{U_i\}$ be an open cover of the n-simpliex Δ^n, then we can barycentrically subdivide Δ finitely many times so that every simpliex is wholy contained in one of the U_i.

Proof. Let $\delta > 0$ be the Lebesgue number for this open cover, then every simplex of diameter $< \delta$ will lie whole in one U_i. So it suffices to show the diameter of the subdivided simplices can be made arbitrarily small with a finite number of subdivisions.

Suppose Δ' is a simplex in the barucentric subdivision of Δ, then I claim:

$$\operatorname{diam}(\Delta') \leq \frac{n}{n+1} \operatorname{diam}(\Delta)$$

which is clearly sufficient. The diameter of a linear simplex in $\mathbb{R}^n$ is in fact equal to the distance between a pair of vertices, so for all pairs of vertices $x, y \in \Delta'$ we just have to consider two cases:

- If neither x nor y is the barycenter b, then they belong to a simplex in the barycentric subdivision of a facet of $\tilde{\Delta} \subset \Delta$, so by induction

$$|x - y| \leq \frac{n-1}{n} \operatorname{diam}(\tilde{\Delta}) \leq \frac{n}{n+1} \operatorname{diam}(\tilde{\Delta}) \leq \frac{n}{n+1} \operatorname{diam}(\Delta)$$

- If instead y is the barycenter b, then if we let b_0 be the barycenter of the facet of Δ opposite $x = v_0$, we have

$$|x - y| = \left| v_0 - \frac{v_0 + \cdots + v_n}{n+1} \right| = \left| \frac{n}{n+1} v_0 - \frac{n}{n+1} \left(\frac{v_1 + \cdots v_n}{n} \right) \right| = \frac{n}{n+1} |v_0 - b_0| \leq \frac{n}{n+1} \operatorname{diam}(\Delta)$$ □

We now just have to put barycentric subdivision into the form of a chain map. We will do this inductively: for a singular n-simplex $\sigma : \Delta^n \to X$ consider each of its facets $\sigma \circ \delta_i$, and consider its barycentric subdivision consisting of $(n-1)$-simplices. Consider the cone of each of these simplices, and send σ to the alternating sums of the subdivision of facets.

$$S(\sigma) = c_b(S(d\sigma))$$